

RESEARCH ARTICLE

Exhaustivity and contrastivity in Hungarian focus constructions: A visual-world study

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Abstract

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Contents

1	Introduction	2
2	Methods	5
2.1	Participants	6
2.2	Materials	6
2.3	Design	8
2.4	Procedure	8
2.5	Analysis	9
3	Results	11
3.1	Mixed-effects modelling of eye-tracking measures	12
4	Discussion	14
5	Conclusion	16

1. Introduction

How languages express contrast and exhaustivity has major implications for how we understand the syntax–semantics interface. This study revisits the debate on the interpretation of Hungarian focus constructions using eye tracking and the visual-world paradigm. Specifically, we examine two theoretically important inquiries: how strongly Hungarian native speakers interpret phrases in the pre-verbal focus domain exhaustively and whether contrastive but non-exhaustive readings exist, as well as from a cognitive perspective, whether different focus types produce processing differences during language comprehension.

- (1) Anna meg-ette a banánt.
Anna[NOM] VPX-eat-PST-3SG.DEF the banana-ACC
'Anna ate (up) the banana.'
- (2) Anna a BANÁNT ette meg.
Anna[NOM] the banana-ACC eat-PST-3SG.DEF VPX
a) 'Anna ate (up) the BANANA.' [and nothing else]
b) 'Anna ate (up) the BANANA.' [and not something else]

Consider the canonical, neutral sentence in example 1, where the subject *Anna* is in the sentence-initial position, followed by the verb, *ette* 'ate' (together with the verbal prefix [VPX] (or verbal modifier [VM]) *meg-*)¹ and the object *banán* 'banana' (with the *-t* accusative ending); vs the pre-verbal focus construction in example 2, where the object is fronted to the pre-verbal position, leaving its (normally attached) verbal prefix behind. The sentence in example 2 is a typical example for a Hungarian focus construction, and it could also be rendered in English with an *it*-cleft, i.e., 'It is the BANANA² that Anna ate.'. The comparison of the Hungarian focus-constructions with English *it*-clefts has its own literature (cf. É. Kiss 1999; Wedgwood et al. 2006), but it is out of the scope of the present discussion.

Constituents in canonical sentences can be focused for various communicative purposes, such as emphasis, question answering, or contradiction. In 2 we are focusing the object, and as such we can observe two changes that mark focus: movement and prosody. Regarding the first, Hungarian has a special, grammaticalized syntactic position for the focused constituent directly in front of the verb – called the pre-verbal focus (PVF) position (Balogh 2007; Wedgwood 2006) or structural focus position (É. Kiss 1998) or simply just focus position – often associated with exhaustive/identificational semantics (Balogh 2013). Besides movement, the syntactically expressed focus is also accompanied by prosodic stress, and it has been suggested that in these focus marking strategies movement is primary, and prosody is secondary (Balogh 2013). Regarding intonation in PVF constructions, the focused constituent typically bears eradicating stress (cf. Gécseg 2020; Kornai & Kálmán 1988), meaning that the verb is destressed and the acoustic prominence of all other elements are leveled. The PVF construction is a common, everyday focus-marking strategy in Hungarian – a topic-prominent, discourse-configurational language where the configuration of constituents expresses information structural roles as opposed to grammatical ones (cf. É. Kiss 1995) – and various aspects of it has garnered significant attention in both theoretical and psycholinguistic research.

¹ Verbal prefixes in Hungarian can modify the meaning of the verb, or act as aspect markers. *Meg-* in this case has a perfective aspect meaning, indicating not only that the action has been concluded but that the event is complete, with the apple wholly eaten.
² Linguistically focused elements in the examples will be set in capitals for highlight throughout the paper.

One of the key issues around Hungarian focus is whether it is inherently exhaustive, or if the exhaustivity is a pragmatic inference. In other words, does the PVF construction itself entail that the focused element is the only one for which the predicate holds true, or can it be interpreted in a contrastive way, where the focused element stands in opposition to other alternatives without necessarily excluding them? For example, in 2 above, does this sentence implies that Anna ate only the banana *and nothing else* – an exhaustive reading, or could it mean that Anna ate the banana *and not something else* (or even, that she ate the banana but maybe *she also ate something else*) – a contrastive reading? We are interested in the ‘default’ interpretation of focus constructions such as these, and whether Hungarian speakers tend to understand them exhaustively or contrastively without additional context, as demonstrated by the two possible readings given in 2.

We should therefore try to define what we mean by *exhaustive focus* and *contrastive focus*. First, both types require the pre-verbal structure, but they differ in their semantic/pragmatic implications. In É.Kiss (1998)’s influential paper, both would fall under the term of *identificational focus*, which she defines as a focus that identifies the referent of the focused element from a set of alternatives. It takes scope, binds a variable, and interacts with quantification. É. Kiss treats contrastive focus as a subtype of identificational focus, where she describes it as [+exhaustive], and often [+contrastive]. However, we will argue that identificational focus does not always entail exhaustivity. According to Rooth (1992)’s theory on focus interpretation, exhaustivity is not inherent to focus but arises when focus interacts with specific operators like ‘only’. For him, focus is a mechanism that introduces a set of alternatives (*alternative semantics*), and he considers contrast to be the pragmatic use of focus, where one alternative is highlighted against others in the set. In Vallduví & Vilkuna (1998), *kontrast* is an operator-like category that introduces a set of alternatives for the focused constituent. If an expression *a* is contrastive, a membership set $M = \{\dots, a, \dots\}$ is generated, serving as a quantificational domain. This allows alternative propositions $P(x)$ to be derived by substituting other members of *M* for *a*. Thus, *kontrast* provides the semantic mechanism by which focus evokes alternatives, enabling identificational or exhaustive readings. In Krifka (2008:259)’s work on Information Structure (IS), both exhaustive and contrastive focus is mentioned as subtypes of focus, and while exhaustive is interpreted as the only alternative that makes the proposition true narrowing the set of alternatives down to one (and thus enforcing exclusivity), contrastive focus is defined as a focus used for truly contrastive utterances and presupposes a competing alternative in the common ground, highlighting the contrast. Krifka’s distinction aligns with his broader distinction between semantic (truth-conditional impact) and pragmatic (discourse management) uses of focus. Zimmermann (2008) also argued that contrastivity is not a semantic property of focus at all but rather a pragmatic one, and that it is best understood in terms of hearer expectation and discourse predictability, as a discourse-semantic device that signals unexpectedness relative to the hearer’s assumptions, rather than a purely semantic operator over alternatives.

In short, exhaustive focus implies that the focused element is the sole entity for which the predicate holds true; i.e. that there are no other things Anna ate. On the other hand, contrastive focus would imply that the focused element stands in opposition to other possible alternatives, drawing attention to its uniqueness, unlikeliness, or contradicting nature given the context, e.g., ‘Anna ate the banana [and not the apple].’, but it does not necessarily exclude alternatives, i.e., there could be other things that Anna ate, e.g., ‘Anna ate the banana [and/but also the apple].’ Now consider the following:

- (3) *Anna csak a BANÁNT ette meg.*
 Anna[NOM] only the banana-ACC eat-PST-3SG.DEF VPX
 ‘Anna ate (up) only the BANANA.’
- (4) *Anna többek között a BANÁNT ette meg.*
 Anna[NOM] others amongst the banana-ACC eat-PST-3SG.DEF VPX
 ‘Anna ate (up), amongst other things, the BANANA.’

We can, in fact, force an exhaustive reading by adding the exhaustive focus marker *csak* ‘only’ to the sentence, as in example 3³. One intuitive reading of a simple PVF sentence like that in example 2 would be that ‘Anna ate the banana [and nothing else].’ In this sense, the sentence seems to be equal to that in 3, where the exhaustive interpretation is explicitly expressed with the operator *csak* ‘only’. If we accept É.Kiss (1998)’s view, this seems to be redundant as the structural focus is already exhaustive. The question of differentiating the exhaustivity entailed by the pre-verbal focus structure vs the exhaustivity operator has been studied in detail by Balogh (2007).

Moreover, we can also cancel exhaustivity within the same clause with the help of certain phrases, such as *többek között* ‘amongst others’, as in example 4⁴, by inserting a phrase with an inherently non-exhaustive, inclusive meaning (see also Balogh 2013; Wedgwood 2006:14). This sentence is not only perfectly grammatical, inclusive of alternatives, but also expresses a contrastive sense due the use of focus (there must be some specific discourse function for choosing to focus a constituent).

In Hungarian linguistic research, the role of semantic exhaustivity in focus constructions has been debated for more than forty years. The central question is whether exhaustivity arises from the syntax–semantics interface or from pragmatic and contextual factors. Earlier accounts within the generative framework assumed that the Hungarian pre-verbal focus structure necessarily entails semantic exhaustiveness, claiming that exhaustivity is an inherent property of the focus structure itself (É.Kiss 1998, 2007; Horvath 2008; Kenei 2006; Szabolcsi 1981), which was more or less the received view up until the 2000s. More recently, this claim has since been challenged and shown to be untrue by both theoretical and psycholinguistic work – supported by experimental data – and it has been proposed that the exhaustive interpretation emerges from discourse-pragmatics, or conversational implicature (Balogh 2013; Káldi and Babarczy 2016; Onea 2009; Wedgwood et al. 2006). In introducing their argument against the claims of Szabolcsi (1981), É.Kiss (1998, 2007), and Horvath (2008) that ‘an immediately pre-verbal focus in Hungarian is semantically exhaustive’ or ‘is interpreted exhaustively, i.e. as if it were in the scope of an exclusive like “only”’, Onea & Beaver (2009:1) suggested in their first example⁵ (analogous to our example 2) that the implicit reading of a sentence of this type of construction should be exhaustive, i.e. ‘Anna ate the banana [and nothing else].’ However, their analysis suggests that while such sentences often invite an exhaustive reading, this interpretation is not structurally encoded but contextually inferred.

Looking at example 2 however, we feel that the underlying implication is not necessarily that Anna ate *nothing* else, but it could be the case that she did not eat *something* else. Therefore, in our view, the interpretation might as well be contrastive.

³The scope of *csak* ‘only’ only extends to the subsequent element, it can only refer to the object and not the following VP.

⁴Similarly, only the object is in the scope of *többek között* ‘amongst others’, it is not possible to interpret this sentence as if the verb is an alternative to other actions one can do to a banana.

⁵*Péter MARIT csókolta meg.* Peter Mari-ACC kiss-PST-3SG.DEF VPX ‘Peter kissed MARY.’

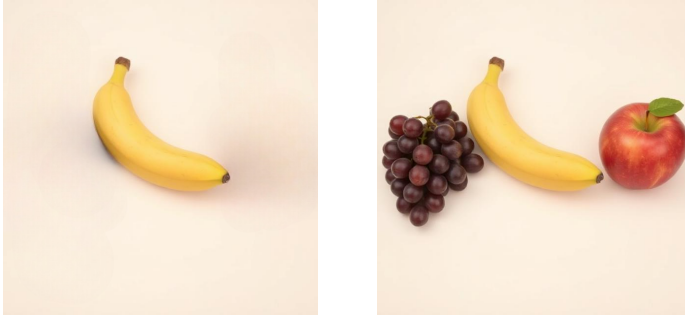


Figure 1. Example stimuli for a sentence–picture verification task

To test this, we designed an experiment within the visual world paradigm where people are first presented with typical PVF sentences, and then given a binary choice that offers two contexts where both are correct and plausible: either A) a single object image corresponding to an exhaustive interpretation, or B) a multi-object image implying a contrastive interpretation (see the image pair in Figure 1 accompanying examples 1–4). Participants have to choose the image that best matches the sentence they just read in a sentence–picture verification task. Their choices should tell us about the preferred interpretation of the PVF constructions, and whether it is more naturally interpreted as exhaustive or contrastive. Besides collecting behavioural data on the choices participants made, we also recorded their reading times and collected eye-tracking data during the picture verification phase, which hopefully can give us additional insights into the cognitive processes involved in interpreting the different types of PVF constructions.

To sum up, while previous work has mainly tried to answer whether an immediately pre-verbal focus in Hungarian is semantically exhaustive or not, our question is: What is the default interpretation of a pre-verbal focus construction? Is it really inherently exhaustive, or could it be **naturally** contrastive?

2. Methods

To address and contribute to the theoretical debate, and in light with recent experimental studies (Gerőcs et al. 2014; Káldi et al. 2021; Onea & Beaver 2009), we designed a visual-world experiment with eye tracking to differentiate interpretations associated with the PVF structures. Specifically, we were curious about Hungarian speakers' dominant interpretation of the unmodified, simple PVF sentence (e.g., example 2), in comparison with the *only*–exhaustive (ex. 3) and the *amongst others*–contrastive conditions (ex. 4), for both of which we had some expectations: *only*–exhaustive sentences should be overwhelmingly interpreted as exhaustive, and *amongst others*–contrastive sentences should probably be interpreted as contrastive. The question was whether the unmodified PVF sentences would pattern with the exhaustive or the contrastive condition, or if they would show a more balanced distribution of interpretations.

2.1. Participants

We recruited 55 native speakers of Hungarian via the Prolific platform (www.prolific.com), where participants were pre-screened for Hungarian as their first language, and received a monetary compensation for their time (around £7.50/h). After rejecting 3 participants (due to overly long submission time, failing attention checks, or missing ET data), the final sample consisted of 52 individuals (32 female; mean age = 35). On average it took people just under 23 minutes to complete the experiment, although with a high individual variation (unusually long submission times were rejected).

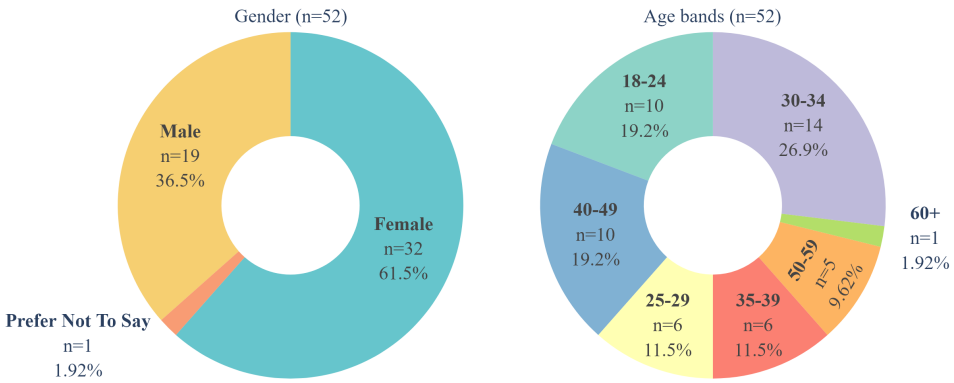


Figure 2. Demographic information of the participants in the experiment; gender (left) and age distributions (right).

2.2. Materials

Our stimuli consisted of 54 experimental sentences (18 items × 3 conditions) with 54 additional filler sentences in a randomized order (108 in total), but making sure that the same items never followed each other. Regarding lexical variation, we used verbs both with and without verbal prefixes (50-50%), and we included instances of both definite and indefinite conjugation (44.4-55.6%); also, both countable and uncountable nouns were used for object focus. We added some simple contextual fillers to the beginning of each sentence, such as *ami X-t illeti* ‘regarding X’ and time or place adverbials to balance sentence length and make the targeted focus constructions appear not in the beginning of trials. When presenting the sentences on the screen, we divided them into regions, and articles were kept together with nouns, so all three conditions had the same number of regions (7), and the focused element was always in region 6 (r6). The stimuli were created by modifying a set of 18 base sentences – which were all simple transitive sentences – by modifying the base sentence with the addition of the focus marker *csak* ‘only’ for the exhaustive condition, and the phrase *többek között* ‘amongst others’ for the contrastive condition.

The 18 items were accompanied by 18 image pairs, these were created using DeepAI's image generator (<https://deepai.org/>) with prompts requesting simple 3D objects & scenarios in front of a light pastel background, and then modified manually to get matching pairs. The full stimuli list (sentences + visuals) is available online ??.

To avoid overly long experimental sessions, we have split the experiment into three equal parts and each participant could complete one, two, or all three parts as separate events; meaning that we had three participant groups that could overlap, and each experiment consisted of 40 trials (36 experimental + 4 attention checks). This resulted in 72 valid submissions (25 for group one, 23 for group two, and 24 for group three), thus every stimuli was interacted with at least 23 times. The three parts were split in a way that for each group each item only appeared in one condition, and each session contained four attention check questions intermixed and evenly distributed across the trials.

The three conditions were: a) exhaustive focus with *csak* 'only' (e.g., 5a), b) unmodified, simple pre-verbal focus (PVF) (e.g., 5b), and c) contrastive focus with *többek között* 'amongst others' (e.g., 5c). Each item was paired with two images: A) one depicting a single-object interpretation – exhaustive reading – and B) one depicting a multiple-object interpretation – contrastive reading. See Figure 3 for an example of the image pairs accompanying the sentences in the experiment.

(5) a. **exhaustive focus**

Ami az innivalót illeti, Eszter csak kávét ivott.
 which[REL] the drink-ACC regard-3SG Eszter only coffee-ACC drink-PST-3SG
 'Regarding drinks, Eszter only drank coffee.'

b. **unmodified pre-verbal focus**

Ami az innivalót illeti, Eszter ma kávét ivott.
 which[REL] the drink-ACC regard-3SG Eszter today coffee-ACC drink-PST-3SG
 'Regarding drinks, Eszter drank coffee today.'

c. **contrastive focus**

Ma reggel Eszter többek között kávét ivott.
 today morning Eszter others amongst coffee-ACC drink-PST-3SG
 'This morning, Eszter drank, amongst other things, coffee.'

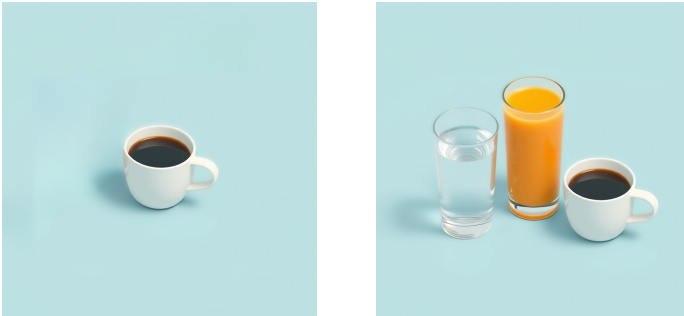


Figure 3. Image pair for the sentence in examples 5a, 5b, & 5c

2.3. Design

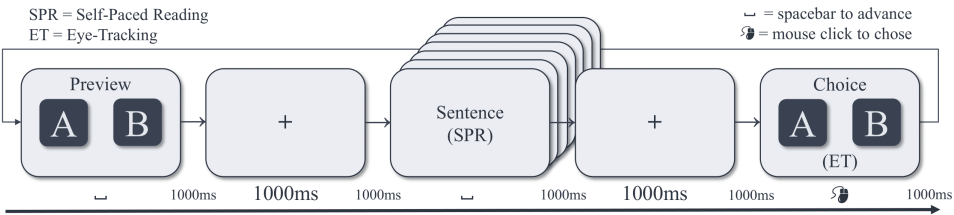


Figure 4. Design

We built a sentence–picture verification (SPV) task, in which participants read a sentence and were asked to indicate one **fitting** meaning-representation between two visual stimuli.

After a practice session consisting of three trials, participants saw a preview of the image pair to get familiar with the objects/context, on the left and right half of the screen (20% of the screen’s width and 40% of the screen’s height) on a canvas 50% larger than the images to mitigate the inaccuracies of the eye tracker. After pressing the spacebar they were presented with the sentence word by word (region by region) in the middle of the screen in a self-paced reading (SPR) manner, in which participants had to use the spacebar to advance through the sentences. We recorded reading times throughout the reading. After reading through the sentence, the image pair re-appeared and participants were asked to select the image that best matched the meaning of the sentence they just read. The choice had to be made with a mouse-click anywhere on the corresponding canvas. Reaction times (RT) and eye movements were recorded during this picture verification phase. The image types were counterbalanced throughout the trials.

2.4. Procedure

The experiment was created, hosted on, and presented to participants on the PCIBex Farm (Schwarz & Zehr 2021), the code and resources are available online ??

Participants completed the experiment on their own devices (PC only, with a webcam and physical keyboard & mouse) and on their own time, but with Prolific helping us control for too fast or too slow submissions resulting in rejection or timeout.

First, potential participants were instructed to be in a well-lit environment, sit at an appropriate distance from their screen and webcam, and minimize head movements; emphasizing the need for a bright environment. Participants then had to calibrate the eye tracker before starting the experiment, which involves looking at specific points on the screen (e.g., corners, center) while the system records their eye positions to create a mapping between gaze coordinates and screen coordinates. Eye-tracking data were collected using the built-in WebGazer.js library of PCIBex, which uses the participant’s webcam to track eye movements, and we followed the default calibration procedure provided (a 9-point calibration grid). Participants had to reach and maintain at least 60% accuracy in the calibration phase to proceed with the experiment, and had three attempts to achieve this accuracy, ensuring that the eye-tracking data collected during the picture verification

phase would be reliable. The tracker also checked gaze before every trial, and if the accuracy dropped below 60%, participants were prompted to recalibrate before proceeding with the next trial. People who could not pass the calibration for various reasons (e.g., computer setup, lighting, environment) were not able to start the experiment. After calibration, participants were instructed on the exact task and how to progress, followed by a practice session, and finally the live experiment.

2.5. Analysis

2.5.1. Eye-tracking data processing

Eye-tracking data recorded on PCIBex comes in a relatively straightforward, clean format not of gaze coordinates but in classified samples, where each sample is already defined as a gaze at a canvas or not, in our case: left canvas, right canvas, or neither. Samples are recorded between every 30-70 milliseconds, depending on the users' computer & browser setup, internet speed, and presumably the general load that the PCIBex server was under at the time.

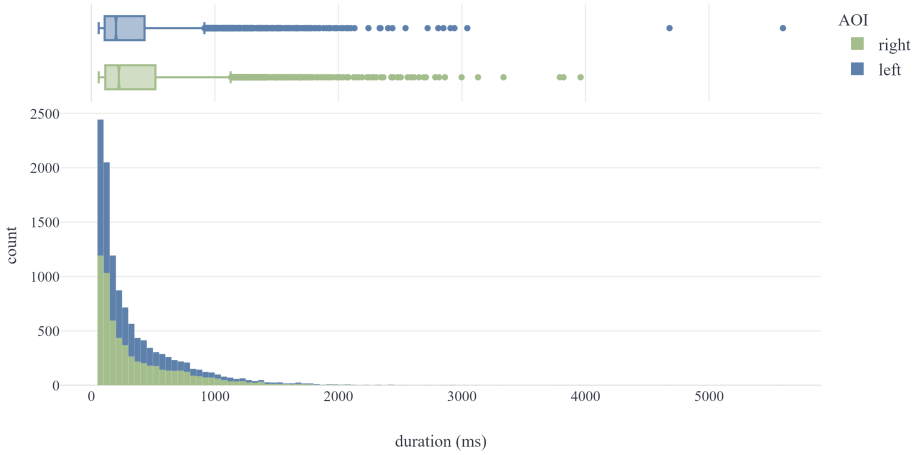


Figure 5. Distribution of episode durations after cleaning and before winsorization.

The following steps were taken to clean and classify the eye-tracking data in order to obtain a certain set of metrics that could be used for analysis: First, raw eye tracking data files were loaded per participant, and **samples** were classified into AOIs at each timestamp as left (left_canvas=1, right_canvas=0), right (right_canvas=1, left_canvas=0), or neutral otherwise. Then, same-side consecutive samples were collapsed into contiguous **episodes** within participant and trial. Episode duration was computed from timestamp-derived sample durations where episodes shorter than a threshold of 60 ms were removed, and neutral-gap bridging was enabled (with a max neutral bridge of 50 ms). In other words, very short gaze episodes and rapid saccades were removed cleaning noise & jitter, as they are unlikely

to reflect meaningful cognitive efforts and more likely to be artifacts of the eye-tracking system. Then, neutral episodes were dropped and excluded from further analysis. Episodes thus intend to represent a moment of sustained gaze within an AOI, ignoring brief out-of-canvas samples that result from technical inaccuracies and are not deliberate (or uncounscious) eye movements. For quality control, episode-duration distributions were inspected via summary statistics, and episodes longer than the mean +2 SD ($\mu \pm 2\sigma$) were winsorized by replacing their duration with the global mean (i.e., with rows retained). This procedure preserved the full number of episodes while limiting the influence of extreme outliers; 4.89% of all episodes were substituted. See Figure 5 for the distribution of episode durations. This cleaning process was devised after established eye-tracking data classification protocols, such as the Identification by Dispersion-Threshold (I-DT) algorithm, which itself is ideal for low- to mid-frequency eye trackers and static stimuli.

Finally, trial-level eye-tracking (ET) features were computed (left/right/total dwell time, dwell proportions, dominant side, fixation counts, revisits, transitions, first-fixation side/duration, and first-visit side/duration). To briefly define the most important metrics: **dwell** is a time value, the sum of the duration of all episodes within a trial (either left, right, or total), representing the total amount of time a participant has spent looking at the AOI(s) during the choice; **fixation** on the other hand is a numeric value, representing how many times a participant looked at the AOIs during a choice, and **transitions** is the number of times a participant switched their gaze from one AOI to the other. Missing trial-level metrics were mean-imputed in 5.02% of the data points, which were mostly due to missing episode-level data (e.g., no episodes detected for a trial, or no episodes remained after cleaning).

We acknowledge that several processing decisions may have limitations/limit inference. Gaze data was essentially collapsed to binary AOIs (left/right), while off-target samples were treated as neutral and excluded. Episodes shorter than 50 ms were removed, potentially excluding very brief but real events. Long-duration episodes were winsorized by replacing values above a global mean+2SD threshold with the global mean, preserving sample size but reducing variance and potentially biasing duration-based measures. Also, missing merged ET trial metrics were mean-imputed for selected variables, which further shrinks the effective sample size. We therefore interpret absolute magnitudes cautiously and focus on relative contrasts between conditions.

The eye-tracking data cleaning and classification pipeline was implemented in Python, and can be inspected [online](#).

2.5.2. Statistical analysis

We analyzed three eye-tracking dependent measures: *total dwell time* (total_dwell), *total fixation count* (total_fixation), and *first-visit duration* (fvd). For each measure, analyses were conducted in R using linear mixed-effects models (packages lme4, car, and emmeans). The fixed-effect predictors were *Condition* (three levels: exclusive, unmodified focus, contrastive) and *Chosen type* (two levels: A, B). To stabilize variance and accommodate zero values, we applied a log transform using $\log(1 + x)$ to the raw dependent measure prior to modeling.

For each dependent measure, we fit the following model with random intercepts for participants and items: $\text{TextM4}y' \sim \text{Condition} \times \text{Choice} + (1 \mid \text{Participant}) + (1 \mid \text{Item})$, where $y' = \log(1 + y)$ is the transformed dependent variable.

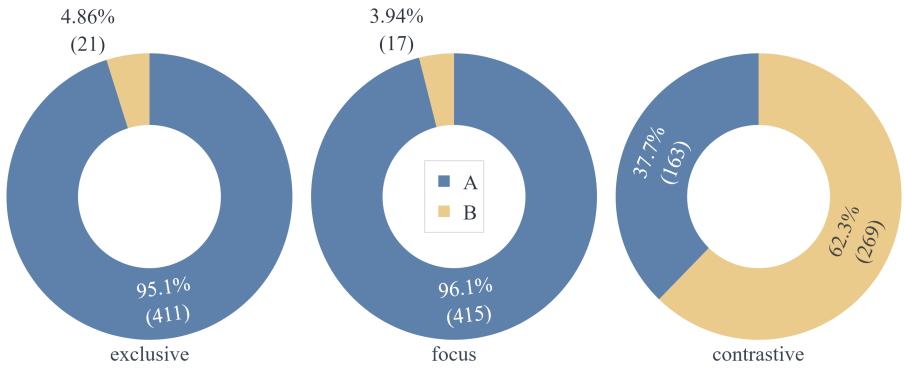


Figure 6. Distribution of chosen visual stimuli (Image type A vs B) per condition.

The model were fit using maximum likelihood estimation (i.e., REML = FALSE) to enable likelihood-ratio model comparisons. For the full interaction model (M4), we report Type-III tests of fixed effects (via `car::Anova`).

To follow up significant effects in M4 – the most promising model – we computed estimated marginal means and conducted Tukey-adjusted pairwise comparisons using `emmeans`. Specifically, we tested (i) pairwise differences between conditions within each type (Condition | Choice) and (ii) pairwise differences between types within each condition (Choice | Condition), with familywise error controlled using Tukey adjustment.

3. Results

The behavioural results showed that participants overwhelmingly interpreted the unmodified pre-verbal focus (PVF) sentences like exhaustive *only*-sentences, choosing single-object type A images (see Figure 6, Table 1). However, in the contrastive condition (*among others*-sentences) people preferred the multi-object type B images, exhibiting a tendency toward non-exhaustive readings, but with almost 37.7% of the choices – somewhat unexpectedly – still being cast on the exhaustive interpretation type A images. This suggests that even with the presence of a phrase that licenses a contrastive reading, the exhaustive effect is still quite dominant. This is clear evidence that the PVF construction is not necessarily interpreted exclusively exhaustively, a contrastive reading can be teased out easily.

Table 1. Type of chosen image per condition..

Condition	Image A: % (N)	Image B: % (N)
Exhaustive	95.1 (411)	4.86 (21)
PVF	96.1 (415)	3.94 (17)
Contrastive	37.7 (163)	62.3 (269)

The reading times across the different regions of the sentences showed no significant differences between conditions, suggesting that the processing of the sentences themselves

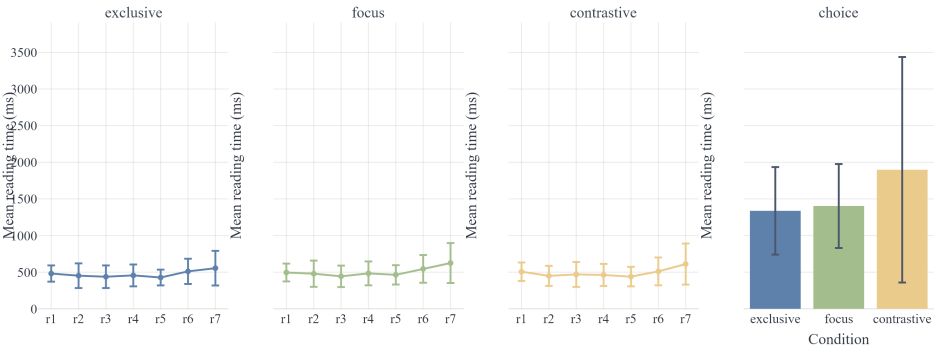


Figure 7. Mean reading times per regions and reaction times of choices.

was not affected by the various focus conditions during self-paced reading. However, a degree of uncertainty and a high level of standard deviation showed up when participants faced with the choice in the contrastive condition, see the reaction times (RT) for the choice in Figure 7.

3.1. Mixed-effects modelling of eye-tracking measures

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To further examine how the focus structures were associated with exhaustive and contrastive readings, we fitted a linear mixed-effects model to (log-transformed) total dwell time on the visual stimuli, with *Condition* (exclusive/only, focus/PVF, contrastive/among others), *Chosen type* (A vs. B), and their interaction as fixed effects, and random intercepts for participant and item. The analysis revealed significant effects of *Condition* ($\chi^2(2) = 12.53, p = .0019$) and *Chosen type* ($\chi^2(1) = 11.85, p = .00058$), qualified by a significant *Condition* $\times$ *Chosen type* interaction ($\chi^2(2) = 12.29, p = .0021$). Post-hoc (Tukey-adjusted) comparisons indicated that when participants ultimately chose type A, dwell times were longer in the contrastive condition than in the exclusive ($\Delta = 0.188, p = .0013$) and focus ($\Delta = 0.140, p = .022$) conditions, while exclusive and focus did not differ ($p = .419$). Comparing choices within each condition, dwell times were shorter for choosing type A than type B in the exclusive condition ($\Delta = -0.444$ for A–B, $p = .0006$), whereas no reliable A–B difference was observed in the focus ($p = .091$) or contrastive ($p = .578$) conditions. Together with the behavioural choice pattern, this suggests a robust exhaustive bias for PVF when no additional contrastive licensing is present, and increased processing cost when an exhaustive (type A) choice is made in a contrastive context.

Until here

Total dwell. A Type-III Wald χ^2 test on the linear mixed-effects model (Model 4) showed significant effects of *condition* ($\chi^2(2) = 12.53, p = .0019$) and *chosen_type* ($\chi^2(1) = 11.85, p = .00058$), qualified by a significant *condition* $\times$ *chosen_type* interaction ($\chi^2(2) = 12.29, p = .0021$). Tukey-adjusted pairwise comparisons of *condition* within *chosen_type* indicated that for type A, *contrastive* yielded longer dwell times than *exclusive* ($p = .0013$) and *focus* ($p = .022$), while *exclusive* and *focus* did not differ ($p = .419$); for type B,

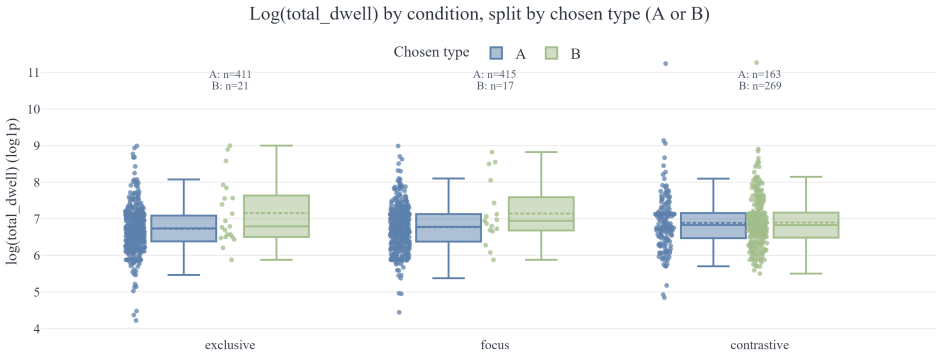


Figure 8. Total dwell time (log-transformed) by condition and choice type.

no condition contrasts were reliable (all $p \geq .069$). Comparing *chosen_type* within each condition showed a reliable type difference only in the *exclusive* condition ($B > A$; $p = .0006$), but not in *focus* ($p = .091$) or *contrastive* ($p = .578$).

Total fixation. For total fixation counts, Model 4 showed a significant main effect of *condition* ($\chi^2(2) = 12.33, p = .0021$) and a significant *condition* $\times$ *chosen_type* interaction ($\chi^2(2) = 6.76, p = .034$), while the main effect of *chosen_type* was marginal ($\chi^2(1) = 3.35, p = .067$). Tukey-adjusted simple effects of *condition* revealed that for type A, *contrastive* produced more fixations than *exclusive* ($p = .0016$) and *focus* ($p = .0107$), with no difference between *exclusive* and *focus* ($p = .687$); for type B, no condition contrasts were significant (all $p \geq .483$). Pairwise comparisons of *chosen_type* within condition were not significant (smallest $p = .068$ in the *exclusive* condition).

FVD. For FVD, Model 4 showed no main effect of *condition* ($\chi^2(2) = 1.65, p = .439$), but a significant main effect of *chosen_type* ($\chi^2(1) = 13.62, p = .00022$) and a significant *condition* $\times$ *chosen_type* interaction ($\chi^2(2) = 9.51, p = .0086$). Tukey-adjusted comparisons of *condition* within *chosen_type* showed no reliable differences for type A (all $p \geq .406$), whereas for type B, *exclusive* elicited higher FVD than *contrastive* ($p = .0056$). Comparisons of *chosen_type* within each condition indicated a robust difference in *exclusive* ($B > A$; $p = .0003$), but not in *focus* ($p = .215$) or *contrastive* ($p = .488$).

END

Table 2. Model-estimated total dwell time during choice (ms; back-transformed from $\log(1+x)$ EMMs). Values are estimated marginal means with standard errors in parentheses..

Condition	Image A (Exhaustive) EMM (SE)	Image B (Contrastive) EMM (SE)
Exhaustive	836 (54)	1299 (181)
PVF	870 (56)	1107 (166)
Contrastive	1002 (75)	972 (67)

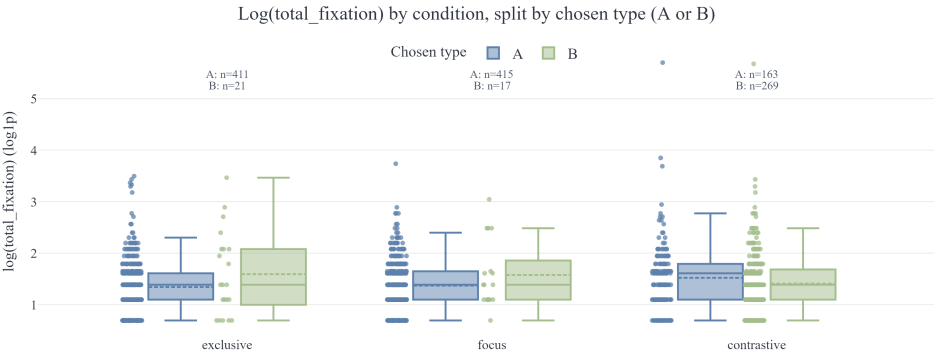


Figure 9. Total fixation count (log-transformed) by condition and choice type.

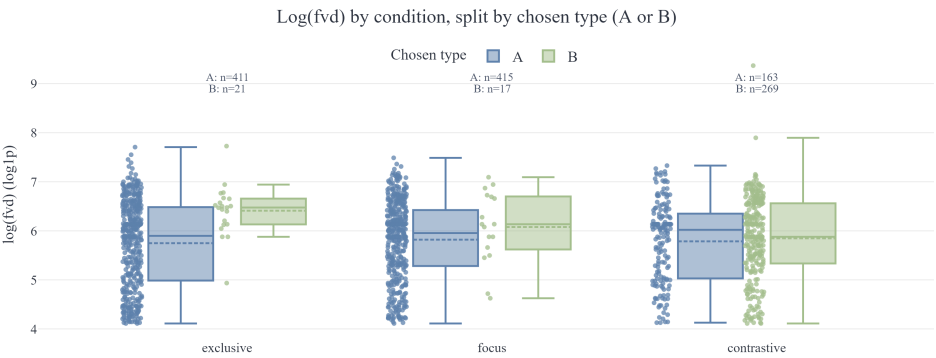


Figure 10. First-visit duration (log-transformed) by condition and choice type.

Table 2 reports model-estimated total dwell time during the choice phase, separated by sentence condition and by the type of image participants selected (A = exhaustive/single-object; B = contrastive/multi-object). In the Exhaustive and PVF conditions, dwell times were lower for exhaustive choices (A) than for contrastive choices (B), whereas in the Contrastive condition dwell times were comparable across the two choice types. Overall, the pattern indicates that verification time depends on the combination of focus condition and the interpretation participants commit to at response.

4. Discussion

Across measures, the results point to a stable but defeasible exhaustive bias for Hungarian pre-verbal focus (PVF). Behaviourally, participants overwhelmingly selected the exhaustive image (type A) for PVF sentences, closely matching explicitly exhaustive (*only*) sentences, while the contrastive *among others* condition shifted choices toward the non-exhaustive image (type B) but did not eliminate exhaustive selections. This pattern already suggests that PVF is not *inherently* contrastive, yet it is also not *categorically* exhaustive in

the way a dedicated exhaustivity marker would be; rather, PVF strongly *invites* an exhaustive interpretation under neutral contexts, while remaining compatible with non-exhaustive continuations when an explicit contrastive (or rather, inclusive) licenser is present.

The eye-tracking measures sharpen this theoretical picture by separating interpretive preferences from processing costs. For total dwell time and total fixation count, we observed a reliable Condition $\times$ Chosen type interaction: when participants ultimately chose the exhaustive image (type A), processing increased in the contrastive condition relative to both *only*/exclusive and PVF. In other words, committing to exhaustivity becomes costlier precisely when the sentence contains an overt cue that makes non-exhaustivity salient (*among others*). By contrast, when participants chose the non-exhaustive image (type B), condition differences were weak or absent. This asymmetry is consistent with a competition account: in a contrastive-licensing context, an exhaustive commitment is still available, but it is harder to justify, leading to additional checking and re-evaluation during the choice phase. The first-visit duration results tell the same story more selectively: condition effects are not global, but appear when the system is pushed into a mismatch (notably, type differences are clearest in the exclusive condition, and the contrastive condition affects type B differently than type A). Taken together, the processing data look less like fixed semantic entailments and more like context-sensitive strengthening with measurable costs when the strengthened reading is disfavoured by discourse cues.

For linguistic theory, this supports approaches in which PVF does not encode exhaustivity as an obligatory semantic operator, but instead yields exhaustivity as a strong default arising from information-structural prominence and pragmatic reasoning (e.g., via Question-under-Discussion alignment, alternative-based competition, or implicature-like strengthening). On this view, *only* supplies explicit exhaustification, producing a relatively coherent mapping to the exhaustive image and (in our data) a robust A–B difference. PVF, in contrast, can be strengthened to exhaustivity in the absence of countervailing cues, explaining the near-ceiling exhaustive choices; crucially, the availability of a contrastive licenser makes the non-exhaustive competitor more prominent, so exhaustive commitments in that environment incur extra processing (longer dwell and more fixations) rather than disappearing. This resolves the apparent tension between accounts that treat PVF as exhaustive and those that emphasize its context dependence: PVF can pattern *as if* exhaustive in neutral contexts, while still permitting non-exhaustive interpretations when the discourse explicitly supports them.

More broadly, the data argue for a graded architecture linking focus, alternatives, and inference: (i) the grammar provides a focus-marked structure that reliably activates alternatives; (ii) discourse cues determine whether strengthening to an exhaustive answer is licensed or costly; and (iii) the cost of sustaining a dispreferred strengthening is visible in online measures even when participants ultimately choose that option. Future work can test this mechanism more directly by manipulating the Question under Discussion and by comparing PVF to other focus strategies (e.g., prosodic focus without movement), which would help localize whether the exhaustive bias is tied to syntactic configuration, prosodic prominence, or the pragmatic profile of the construction.

Tie all this back to the literature!!!

5. Conclusion

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Unnumbered section

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Data availability statement. Replication data and code can be found at <https://doi.org/link>.

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